

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 7 - Summative Assessment 1

Summative Assessment (SA-1) - Answer Key

Class: Class 7

Assessment Type: Summative Assessment (SA-1)

Total Marks: 40

Duration: 2 hours

Topics Covered: Ch 1: Networking & Internet, Ch 2: Advanced Graphics, Ch 3: Multimedia,
Ch 4: Data Handling

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Multiple Choice Questions (10 marks)

Q.No	Answer	Explanation
1	c) MAN	MAN (Metropolitan Area Network) spans across a city.
2	b) HTTPS	HTTPS (HyperText Transfer Protocol Secure) provides secure web browsing with encryption.
3	a) Uniform Resource Locator	URL stands for Uniform Resource Locator, the address of a web page.
4	b) Rectangle Select Tool	The Rectangle Select Tool is used to select rectangular areas in GIMP.
5	c) .mp3	.mp3 is the most common audio format for music files.
6	a) Add values from A1 to A5	=SUM(A1:A5) adds all numeric values in cells A1 through A5.
7	b) Inkscape	Inkscape is a vector graphics editor. GIMP is raster, Audacity is audio, VLC is media player.
8	b) To protect a network from unauthorized access	A firewall monitors and controls incoming and outgoing network traffic.
9	b) Automatically calculates the sum of selected cells	AutoSum quickly adds selected cells using the SUM function.
10	c) Video	Video deals with moving pictures combined with sound.

Part B: Short Answer Questions (16 marks)

- Differentiate between LAN and WAN with one example each.**
 - Answer:** LAN connects devices within a small area like a building. WAN covers large geographical areas like countries. Example of LAN: School computer lab. Example of WAN: Internet connecting computers worldwide.
 - Marking:** 1 mark for LAN, 1 mark for WAN with examples.
- What is a search engine? Name any two search engines.**
 - Answer:** A search engine is a software system that searches the Internet for information based on user queries. Two search engines: Google, Bing.
 - Marking:** 1 mark for definition, 1 mark for names.
- Explain the use of layers in GIMP. Why are they important for image editing?**
 - Answer:** Layers are transparent sheets on which different parts of an image are placed. They are important because they allow non-destructive editing, enable editing one part without affecting others, and make it easy to rearrange elements.

- **Marking:** 1 mark for definition, 1 mark for importance.
4. **What is the difference between raster and vector graphics? Give one example of each.**
 - **Answer:** Raster graphics are made of pixels and have fixed resolution (e.g., JPEG photos). Vector graphics use mathematical paths and scale without quality loss (e.g., SVG logos).
 - **Marking:** 1 mark for each type with example.
 5. **Define multimedia. List its five main components.**
 - **Answer:** Multimedia is the integration of multiple media forms. Components: Text, Audio, Video, Graphics, Animation.
 - **Marking:** 1 mark for definition, 1 mark for components.
 6. **Write the syntax of the IF function in LibreOffice Calc with a practical example.**
 - **Answer:** Syntax: =IF(condition; value_if_true; value_if_false). Example: =IF(A1>=60; "Pass"; "Fail").
 - **Marking:** 1 mark for syntax, 1 mark for example.
 7. **What is the difference between a formula and a function in a spreadsheet?**
 - **Answer:** A formula is a user-defined expression using operators (e.g., =A1+B1). A function is a predefined formula provided by the software (e.g., SUM, AVERAGE).
 - **Marking:** 1 mark for formula, 1 mark for function.
 8. **What are the advantages of using charts in data analysis?**
 - **Answer:** Charts help visualize data trends, make comparisons easier, identify patterns, present data clearly to audiences, and simplify complex data sets.
 - **Marking:** 0.4 marks for each advantage (2 marks for five advantages).

Part C: Long Answer Questions (12 marks)

1. **Explain the types of computer networks (LAN, MAN, WAN) with a neat diagram. Compare their features, advantages, and disadvantages.**
 - **Answer:**
 - **LAN:** Small area, high speed, low cost. Advantages: Resource sharing, easy maintenance. Disadvantages: Limited range, security risks if not configured.
 - **MAN:** City-wide, moderate speed, higher cost. Advantages: Connects multiple LANs, covers larger area. Disadvantages: More complex setup, higher maintenance.
 - **WAN:** Global, lower speed, highest cost. Advantages: Global connectivity, unlimited range. Disadvantages: Expensive, slower speeds, security concerns.
 - **Marking:** 1 mark for each definition, 1 mark for features comparison, 1 mark for diagram.
2. **Describe the steps to edit an image in GIMP using layers. Explain how to add a new layer, draw on it, and merge layers.**
 - **Answer:**
 - Step 1: Open image in GIMP.
 - Step 2: Go to Layer menu → New Layer to add a new layer.
 - Step 3: Select the new layer and use drawing tools to add content.
 - Step 4: Use Layer menu → Merge Down to combine layers.
 - Step 5: Save the edited image.
 - **Marking:** 2 marks for steps, 1 mark for explaining merge process.
3. **What is multimedia? Explain the role of audio and video in multimedia with examples. How is compression used in multimedia?**
 - **Answer:**
 - Multimedia combines text, audio, video, graphics, and animation.

- **Audio role:** Adds sound effects, narration, and music (e.g., background music in presentations).
 - **Video role:** Provides visual storytelling and demonstrations (e.g., educational videos).
 - **Compression:** Reduces file size for storage and transmission (e.g., MP3 for audio, MP4 for video).
 - **Marking:** 1 mark for definition, 1 mark for audio and video roles, 1 mark for compression.
4. **Explain the components of a spreadsheet with a suitable example. Describe how to use SUM, AVERAGE, MAX, and MIN functions.**
- **Answer:**
 - Components: Cells, rows, columns, sheets, formulas, functions.
 - **SUM:** Adds values (e.g., =SUM(A1:A10)).
 - **AVERAGE:** Finds mean value (e.g., =AVERAGE(B1:B10)).
 - **MAX:** Returns largest value (e.g., =MAX(C1:C10)).
 - **MIN:** Returns smallest value (e.g., =MIN(D1:D10)).
 - **Marking:** 1 mark for components, 0.5 marks for each function.

Part D: Application Based Questions (2 marks)

1. **Raj wants to create a photo album with multiple images. Which image editing software would you recommend and why? Mention two features that would help him.**
- **Answer:** Recommend GIMP because it is free and supports layers. Features: 1) Layer support for organizing multiple images. 2) Filters and effects to enhance photos.
 - **Marking:** 1 mark for recommendation, 1 mark for features.
2. **A school wants to connect computers in 5 different buildings across a campus. Which type of network would be most suitable? Justify your answer.**
- **Answer:** MAN (Metropolitan Area Network) is most suitable because it covers a campus area larger than a single building but smaller than a city. It provides high-speed connectivity between buildings.
 - **Marking:** 1 mark for correct network type, 1 mark for justification.